

Notes: Cole, Paulson, and Shastry (RFS, 2014)

Smart Money? The Effect of Education on Financial Outcomes

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1 Big picture

- **Question.** Does general education causally improve household financial outcomes, or is the observed education–finance correlation mainly driven by unobserved ability, family background, income, or selection?
- **Core empirical challenge.** More educated households tend to have more wealth, better jobs, higher cognitive skills, and different family backgrounds. A simple regression of financial outcomes on schooling therefore does not isolate the causal effect of schooling.
- **Main design.** The paper uses historical variation in U.S. state compulsory schooling laws as instruments for educational attainment. These laws change how long students are required to remain in school. The identifying idea is that changes in these laws shift schooling for some marginal students but are not otherwise chosen by individual households.
- **Main population.** The estimates are local average treatment effects for people whose final schooling was changed by compulsory schooling laws. This is not the effect of college for college-bound students. It is mainly the effect of additional schooling for people near the high-school-completion margin.
- **Asset-market result.** One additional year of schooling raises the probability of reporting any investment income by about 7–8 percentage points. It also raises direct ownership of higher-return assets: the IV estimates imply increases of about 6.5 percentage points for bonds or government securities and about 4 percentage points for stocks or mutual funds.
- **Retirement and intensive-margin result.** More schooling raises the probability of reporting retirement income and raises the dollar amount of investment and retirement income. The estimated effect on investment income is about \$1,762 per additional year of schooling.
- **Credit-market result.** Using a two-sample IV design that combines Census first-stage estimates with the FRBNY Consumer Credit Panel/Equifax data, the paper estimates that one more year of schooling reduces bankruptcy by about 3.3 percentage points, reduces foreclosure by about 5.7 percentage points, raises credit scores by about 7.7 points, and reduces delinquency.
- **Crisis result.** The foreclosure effect is especially strong in the 2007–2011 crisis/postcrisis period. Education appears to reduce exposure to the most severe household financial mistakes when financial decisions become especially costly.
- **Mechanism.** Higher labor income alone does not explain the results. A back-of-the-envelope calibration says that the wage return to education would have to be implausibly large to generate the observed increase in investment income if savings behavior did not also change. The evidence points toward better savings, investment, and credit-management behavior.
- **Broader takeaway.** Education is not only a labor-market investment. It also changes household financial management. Counting only wage gains likely understates the private and social returns to education.

2 Data and measurement (key objects)

2.1 Three datasets and why each is used

The paper combines three complementary datasets:

- **U.S. Census, 1980, 1990, and 2000.** A 5% public-use sample with roughly 14.9 million observations. It has very large sample size and allows precise IV estimation, but it has limited financial information.
- **Survey of Income and Program Participation (SIPP), 1984–2008.** A national panel survey with direct information on financial asset ownership. It is smaller than the Census but lets the authors validate whether Census investment income is a good proxy for financial market participation.
- **FRBNY Consumer Credit Panel/Equifax (FRBNY-CCP), 1999–2011.** A quarterly credit-bureau panel. It contains credit scores, bankruptcy, foreclosure, and credit-card delinquency outcomes, but it does not report education.

Interpretation. The Census is powerful for education-IV estimation and asset-income outcomes; the SIPP confirms the asset-participation interpretation; the credit bureau data extend the analysis from asset accumulation to credit management.

2.2 Census outcomes

The Census does not report detailed balance sheets or portfolios. The paper therefore focuses on income generated by financial assets.

The main Census outcome is **investment income**, defined as income from interest, dividends, net rental income, royalties, estates, and trusts. The key binary variable equals one if a household reports any nonzero investment income, positive or negative.

Other Census outcomes include:

- an indicator for investment income greater than \$500 or negative;
- indicators for investment income exceeding predicted savings-account interest thresholds based on the SCF or SIPP;
- the dollar amount of investment income;
- an indicator for any retirement income;
- the dollar amount of retirement income.

Interpretation. The binary measure is designed to capture whether a household has any return-yielding financial assets. It is imperfect because the Census does not directly report portfolios, but it is informative in a very large sample.

2.3 SIPP outcomes

The SIPP provides direct indicators for whether an individual has:

- any transaction account;

- any bonds or government securities;
- any stocks or mutual funds.

Interpretation. The SIPP is useful because it separates basic account ownership from participation in higher-return financial markets. A key result is that schooling does not materially affect transaction-account ownership, but it does affect bonds and stocks.

2.4 FRBNY-CCP credit outcomes

The credit-bureau sample is a 5% panel of American borrowers, restricted to people with data in every quarter from 1999 to 2011. The main outcomes are:

- **Bankruptcy indicator:** whether the individual declared bankruptcy at least once between 1992 and 2011;
- **Foreclosure indicator:** whether the individual experienced foreclosure at least once between 1992 and 2011;
- **Credit score:** a credit-risk score similar to a FICO score, averaged across quarters;
- **Percent balance current:** nondelinquent credit-card balance divided by total credit-card balance;
- **Percent quarters delinquent:** share of quarters in which the borrower had any delinquent credit-card balance.

Interpretation. These outcomes capture financial distress and credit management rather than asset accumulation. Bankruptcy and foreclosure are especially severe outcomes with large private and social costs.

2.5 The instrument: compulsory schooling laws

The instrumental variables are indicators for the amount of schooling required by the law in the individual's state when the individual turned age 14. The omitted category is a law requiring eight or fewer years of schooling. The included instruments are:

$$Comp9_i, \quad Comp10_i, \quad Comp11_i, \quad (1)$$

where each variable indicates whether the state required schooling through grade 9, grade 10, or grade 11 or more.

Interpretation. The instrument changes the minimum legal schooling level. It is meant to affect financial outcomes only by changing actual education, not by directly changing financial behavior.

2.6 Important descriptive facts

From Table 1:

- average years of schooling in the Census sample is 12.91;
- 28.9% of Census households report any nonzero investment income;

- average investment income is about \$1,811;
- 21.9% of SIPP respondents own stocks or mutual funds;
- in the credit-bureau sample, 14.4% experience bankruptcy and 5.8% experience foreclosure over the relevant window;
- the average credit score is about 715.

Interpretation. The outcomes have meaningful baseline rates. A 4–8 percentage point education effect on participation or a 3–6 percentage point effect on bankruptcy/foreclosure is economically large relative to these means.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline structural equation

The paper’s core empirical object is the causal effect of schooling on a financial outcome:

$$y_i = \alpha + \beta s_i + \gamma' X_i + \varepsilon_i. \quad (2)$$

Here, y_i is a financial outcome, s_i is years of education, and X_i includes controls such as age, gender, race, state of birth, state of residence, survey year, birth-cohort fixed effects, and, in many specifications, a flexible control for earned income.

Interpretation. The coefficient β is the object of interest: how one additional year of schooling changes the financial outcome. Ordinary least squares is not enough because s_i is correlated with ability, family background, preferences, and other unobservables.

3.2 First-stage equation

The first stage uses compulsory schooling requirements to predict education:

$$s_i = \alpha + \delta_9 \text{Comp9}_i + \delta_{10} \text{Comp10}_i + \delta_{11} \text{Comp11}_i + \gamma' X_i + \nu_i. \quad (3)$$

The omitted category is a compulsory schooling rule requiring eight or fewer years. The coefficients $\delta_9, \delta_{10}, \delta_{11}$ measure how much more schooling individuals receive when they are exposed to stricter schooling laws.

Interpretation. A strong first stage means the law actually moved educational attainment. In the Census, the instruments are strong: the excluded-instrument F-statistic for years of schooling is 47.2.

3.3 Two-stage least squares for asset outcomes

The IV estimate replaces actual schooling with predicted schooling from the first stage:

$$y_i = \alpha + \beta \hat{s}_i + \gamma' X_i + u_i. \quad (4)$$

The outcome y_i is usually an investment-income, retirement-income, or asset-ownership variable.

Interpretation. The IV estimate is a LATE. It answers: for people whose schooling was shifted by compulsory schooling laws, how did additional schooling change their financial outcomes?

3.4 Reduced form for credit outcomes

Because the credit bureau data do not contain education, the paper first estimates the direct effect of compulsory schooling laws on credit outcomes:

$$y_i = \alpha + \beta_9 Comp9_i + \beta_{10} Comp10_i + \beta_{11} Comp11_i + \gamma' X_i + \varepsilon_i. \quad (5)$$

The paper also estimates a version with a single variable for years of compulsory schooling:

$$y_i = \alpha + \rho CompYears_i + \gamma' X_i + \varepsilon_i. \quad (6)$$

Interpretation. The reduced form is policy-relevant. It tells us what happens to credit outcomes when the legal schooling requirement changes, including both people whose schooling changes and people whose schooling would have been the same anyway.

3.5 Two-sample IV for credit outcomes

The credit bureau data do not report education, so the paper uses a two-sample IV method.

Step 1: estimate the education first stage in the Census:

$$s_i = \alpha + \delta_9 Comp9_i + \delta_{10} Comp10_i + \delta_{11} Comp11_i + \gamma' X_i + \nu_i. \quad (7)$$

Step 2: use the estimated first-stage coefficients to construct predicted schooling in the credit bureau data:

$$\hat{s}_j = \hat{\alpha} + \hat{\delta}_9 Comp9_j + \hat{\delta}_{10} Comp10_j + \hat{\delta}_{11} Comp11_j + \hat{\gamma}' X_j. \quad (8)$$

Step 3: regress credit outcomes on predicted schooling:

$$y_j = \alpha + \theta \hat{s}_j + \Gamma' X_j + \eta_j. \quad (9)$$

Interpretation. This design lets the authors estimate the effect of education on credit outcomes even though education is not observed in the credit data. The cost is that inference is more complicated, so the paper reports Murphy–Topel standard errors and block-bootstrap standard errors.

3.6 Precrisis versus crisis specifications

For bankruptcy and foreclosure, the paper also splits the credit outcomes into:

- a precrisis period, 1999Q2–2007Q3;
- a crisis/postcrisis period, 2007Q3–2011Q4.

Interpretation. This split asks whether education matters more when bad financial decisions are more likely to trigger severe losses. The answer is yes for foreclosure: the estimated schooling effect is larger in the crisis/postcrisis period.

4 Identification and instruments (what makes the estimates credible)

4.1 Relevance: compulsory schooling laws move education

Table 3 shows that stricter compulsory attendance laws raise schooling. In the Census first stage:

- requiring attendance through grade 9 raises schooling by about 0.214 years;
- requiring attendance through grade 10 raises schooling by about 0.199 years;
- requiring attendance through grade 11 or more raises schooling by about 0.266 years;
- the F-statistic for the excluded instruments is 47.2 for years of schooling.

Interpretation. These laws matter for educational attainment, especially for people near the dropout margin. That is the first requirement for the IV strategy.

4.2 Exclusion restriction

The identifying assumption is that, conditional on controls, the compulsory schooling law in force when a person turned 14 affects later financial outcomes only through schooling.

The control set is rich. It includes age-group effects, birth-cohort effects, state of birth, state of residence, survey year, gender, race, and, in many specifications, flexible earned-income controls. The paper also follows Lochner and Moretti by including interactions designed to absorb the effect of *Brown v. Board of Education* on southern cohorts.

Interpretation. The design compares people born in different states and cohorts who faced different compulsory schooling rules, after removing broad state, cohort, residence, demographic, and historical patterns.

4.3 Why the estimate is a LATE

The laws do not change schooling for everyone. Many people would stay in school regardless of the law; others may drop out despite the law. The IV estimate identifies the effect for the group whose schooling is changed by the law.

Interpretation. This is a useful population for policy. It is likely to include lower-education and lower-income individuals whose financial outcomes are especially relevant for household finance and consumer protection.

4.4 Role of income controls

Some specifications include a cubic polynomial in earned income or flexible income controls.

This is not needed for identification in the pure IV sense, and income may itself be affected by schooling. But it is an informative mechanism check: if the education coefficient remains large after conditioning flexibly on earnings, then education is unlikely to operate only through higher wages.

Interpretation. The estimates remain large with income controls. This supports the view that education changes financial behavior, not only labor income.

4.5 Two-sample IV assumptions for credit outcomes

The credit bureau data do not report education, state of birth, race, gender, or income. The paper therefore:

- uses Census data to estimate the first stage;
- uses state of residence in the first quarter of the credit panel as a proxy for state of birth;
- uses ZIP-code fixed effects in some reduced-form specifications to absorb local income and demographic differences;
- reports both Murphy–Topel and bootstrap standard errors.

Interpretation. This part of the paper is more indirect than the Census/SIPP analysis. The gain is that it permits a causal education design for rare, severe credit outcomes in a very large sample.

4.6 Main threats and how the paper addresses them

- **Omitted ability or family background.** The IV strategy is meant to break the link between individual unobservables and education.
- **State-level trends.** State-of-birth, state-of-residence, year, cohort, and Brown v. Board controls absorb many broad confounds.
- **Investment income as a proxy.** The paper uses alternative thresholds and SIPP direct asset ownership to validate the interpretation.
- **Credit-data missing education.** The two-sample IV approach uses a first stage from the Census and adjusts inference for generated regressors.
- **Migration.** State of birth is ideal for the schooling-law assignment. In the credit data, state of residence is only a proxy, likely adding noise and attenuating estimates if migration is unrelated to the outcome conditional on controls.

Interpretation. No single test proves the exclusion restriction. The credibility comes from the combination of a historical policy instrument, strong first stage, rich controls, robust asset-outcome definitions, and consistent patterns across independent datasets.

5 Tables and figures

The paper is mainly table-based. The most important exhibits are Tables 1–6. The images below are included as visual anchors; the main interpretation is in the **Read** and **Economic message** paragraphs.

5.1 Table 1: Summary statistics

Read:Table 1 describes the main samples and outcome means. The Census sample has about 14.9 million individuals, average schooling of 12.91 years, and a 28.9% rate of any investment income. In the SIPP, 21.9% own stocks or mutual funds. In the credit-bureau sample, 14.4% experience bankruptcy, 5.8% experience foreclosure, the average credit score is 714.7, and the average share of quarters with credit-card delinquency is 7.5%.

Economic message. These means make the magnitudes easy to read. For example, a 4 percentage point increase in stock ownership or a 5.7 percentage point reduction in foreclosure is not a small effect relative to the baseline.

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Table 1
Summary statistics

	Mean	SD	N
Panel A: Household survey data			
Demographics (census)			
Age	45.60	(14.74)	14,913,356
Years of schooling	12.91	(2.69)	14,913,356
Compulsory attendance ≤ 8	0.153	(0.360)	14,913,356
Compulsory attendance = 9	0.404	(0.491)	14,913,356
Compulsory attendance = 10	0.101	(0.301)	14,913,356
Compulsory attendance = 11	0.342	(0.474)	14,913,356
Income from investments (census)			
Indicator for nonzero income	0.289	(0.453)	14,913,356
Indicator: Income > 500 or negative	0.189	(0.391)	14,913,356
Indicator: Income > predicted savings	0.278	(0.448)	14,913,356
Indicator: Income > predicted savings	0.296	(0.456)	4,053,909
account interest (SIPP)* or negative			
Amount	1,810.62	(9,250.44)	14,913,356
Income from retirement savings (census)			
Indicator: Income > 0	0.224	(0.417)	4,150,828
Amount	3,315.67	(10,635.99)	4,150,828
Transactions account (SIPP)			
Indicator for having	0.766	(0.423)	168,946
Bonds or government securities (SIPP)			
Indicator for having	0.148	(0.355)	262,245
Stocks or mutual funds (SIPP)			
Indicator for having	0.219	(0.413)	270,316
Panel B: Credit bureau data			
Bankruptcy indicator	0.144	(0.351)	5,750,005
Foreclosure indicator	0.058	(0.234)	5,750,005
Credit score	714.67	(90.57)	5,732,690
% Balance current	0.956	(0.113)	5,329,619
% Quarters delinquent	0.075	(0.152)	5,750,005

This table reports summary statistics for data used in this paper. Panel A reports summary statistics from the 5% sample of the census (1980, 1990, and 2000) as well as various SIPP waves (1984–2008). Indicators for having bonds or government securities and stocks or mutual funds are from all 1984–2008 SIPP waves, whereas the indicator for having a transaction account is from the 1990–2008 SIPP waves. Panel B reports summary statistics for data from the FRBNY Consumer Credit Panel/Equifax. The sample comprises a 5% panel of American borrowers, restricted to borrowers who have data in every quarter of the panel from 1999 to 2011. Bankruptcy and Foreclosure are indicators for having undergone bankruptcy or foreclosure at least once, respectively, between 1992 and 2011. Credit Score is averaged for each individual across all quarters of data, and it can range from

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Table 2
OLS estimates of the effect of years of schooling on income from various sources

Panel A: Census outcomes				
	Indicator: Any income from investments	Amount of income from investments	Indicator: Any income from retirement savings	Amount of income from retirement savings
	(1)	(2)	(3)	(4)
Years of schooling	0.035*** (0.0001)	271.55*** (5.02)	0.024*** (0.0002)	548.42*** (4.81)
No. of observations	14,913,356	14,838,407	4,150,828	4,117,987
R ²	0.184	0.092	0.177	0.147
Panel B: SIPP outcomes				
	Indicator: Any transactions account	Indicator: Any bond or government securities	Indicator: Any stocks or mutual funds	
	(1)	(2)	(3)	
Years of schooling	0.026*** (0.0005)	0.0162*** (0.0003)	0.029*** (0.0003)	
No. of observations	168,946	262,245	270,316	
R ²	0.133	0.064	0.128	

This table reports results from regressions of income and assets on years of schooling, gender, race, age (3-year age groups), birth cohort (10-year cohorts), state of birth, state of residence, survey year, and a cubic polynomial in earned income. Only the coefficient on education is reported. Regressions also include state of birth fixed effects interacted with a dummy variable for being born in the South and turning age 14 in 1958 or later to account for the impact of *Brown v. Board of Education*. We include 18- to 75-year-olds (50- to 75-year-olds when considering retirement income). In Panel A, the sample comprises individuals reported in the 5% samples of the 1980, 1990, and 2000 censuses. The dependent variable of interest is whether the household receives income from investments or retirement savings and the amount. In Panel B, the sample comprises individuals from the

5.2 Table 2: OLS relationships

Read:The OLS estimates show the familiar positive correlation between schooling and financial outcomes. Each additional year of schooling is associated with a 3.5 percentage point increase in any investment income, about \$272 more investment income, a 2.4 percentage point increase in retirement-income receipt, and a 2.9 percentage point increase in stocks or mutual fund ownership.

Economic message. OLS confirms that education and financial behavior are strongly related, but it is not causal because education is correlated with ability, wealth, parental background, patience, and many other variables. Table 2 is a benchmark, not the main evidence.

5.3 Table 3: First stage

Read:In the Census, requiring more years of school raises actual education and high-school completion. The estimates for years of schooling are about 0.214 for grade 9, 0.199 for grade 10, and 0.266 for grade 11 or more, all relative to the omitted category of eight or fewer years required. The F-statistic for years of schooling is 47.2 in the Census and 11.5 in the SIPP.

Economic message. The first stage is strong enough to support IV estimation. It also clarifies the economics of the LATE: the law shifts schooling by fractions of a year because only marginal students are induced to stay longer.

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Table 3
Estimates of the effect of compulsory schooling laws on education

	Years of schooling (1)	High school (2)	Years of schooling (3)	High school (4)
Compulsory attendance = 9	0.214*** (0.018)	0.039*** (0.003)	0.0260 (0.034)	0.0120** (0.005)
Compulsory attendance = 10	0.199*** (0.024)	0.041*** (0.004)	0.1660*** (0.046)	0.0265*** (0.007)
Compulsory attendance = 11	0.266*** (0.028)	0.055*** (0.005)	0.1747*** (0.040)	0.0428*** (0.006)
No. of observations	14,913,356	14,913,356	276,079	276,079
R ²	0.234	0.178	0.182	0.137
Data source	census	census	SIPP	SIPP
F-stat of excluded instruments	47.2	52.4	11.5	20.8

This table reports the first-stage relationship between compulsory school laws and educational attainment. In Columns 1 and 2, the sample comprises individuals reported in the 5% samples of the 1980, 1990, and 2000 censuses. In Columns 3 and 4, the sample comprises individuals from the 1984–2008 SIPP waves. We include 18- to 75-year-olds. The dependent variables of interest are the number of years of schooling attained (Columns 1 and 3) and an indicator for whether the individual graduated high school (Columns 2 and 4). The independent variables of interest indicate whether the state in which the individual was born prohibited dropout until a child had completed 9th, 10th, or 11th grade and higher (requiring 8 or fewer years of schooling is the omitted category). Other controls include fixed effects for gender, race, 3-year age groups, 10-year birth cohorts, state of birth, state of residence, survey year, and a cubic polynomial in earned income. Regressions also include state of birth fixed effects interacted with a dummy variable for being born in the South and turning age 14 in 1958 or later, to

Table 4
IV estimates of the effect of years of schooling on income from various sources

Panel A: Census outcomes

	Indicator: Any income from investments (1)	Indicator: Income from investments > \$500 (2)	Indicator: Income from investment > "cutoff" (SCF) (3)	Indicator: Income from investment > "cutoff" (SIPP) (4)	Amount of income from investments (5)	Indicator: Any income from retirement savings (6)	Amount of income from retirement savings (7)
Years of schooling	0.060*** (0.005)	0.075*** (0.005)	0.066*** (0.007)	0.161*** (0.019)	0.096*** (0.016)	1761.58*** (128.59)	0.059*** (0.010)
No. of observations	14,913,356	14,913,356	14,913,356	4,683,809	14,838,407	4,150,828	4,117,987
F-stat of excluded instruments	57.7	47.2	47.2	9.08	47.1	45.0	44.5
Cubic control for income	no	yes	yes	yes	yes	yes	yes

Panel B: SIPP Outcomes

	Indicator: Any transactions account (1)	Indicator: Any bond or government securities (2)	Indicator: Any stocks or mutual funds (3)	Indicator: Any stocks or mutual funds (4)
Years of schooling	-0.002 (0.031)	0.002 (0.027)	0.0686*** (0.020)	0.0686*** (0.020)
No. of observations	171,361	168,046	265,173	262,245
F-stat of excluded instruments	8.416	11.485	8.416	11.485
Cubic control for income	no	yes	no	yes

This table reports results from 258.5 regressions of income and assets on years of schooling, gender, race, age (3-year age groups), birth cohort (10-year cohorts), state of birth, state of residence, survey year, and (where noted) a cubic polynomial in earned income. Only the education coefficient is reported. The first stage is a regression of years of schooling on compulsory attendance demands for grades 9, 10, and 11 or more (requiring 8 or fewer years is the omitted category). Regressions also include state of birth fixed effects interacted with a dummy variable for being born in the South and turning age 14 in 1958 or later to account for the impact of *Brown v. Board of Education*. We include 18- to 75-year-olds (50- to 75-year-olds who considering retirement income). In Panel A, the sample comprises individuals reported in the 5% samples of the 1980, 1990, and 2000 censuses. The dependent variable of interest is whether the household receives income from investments or retirement savings and the amount. In Column 5, "Cutoff (SCF)" is predicted from an OLS regression of other interest income on age, race, sex, earned income, and year indicators, where the sample includes households from the 1983, 1992, and 2001 SCF with other interest income less than \$50,000. In Column 5, "Cut (SIPP)" is predicted from an OLS regression of savings account interest on age, earned income, race, sex, and state of residence indicators, where the sample includes individuals from 8 2001 SIPP. In Panel B, the sample comprises individuals from the 1990–2008 SIPP waves (Columns 1 and 2) and the 1984–2008 SIPP waves (Columns 3–6). In Panel A, Columns 6 and 8, top-coded individuals (see text) are dropped. Standard errors, corrected for arbitrary correlation within state of birth-year of birth, are in parentheses. Numbers with * or *** indicate significance at the 10%, or 5%, level, respectively.

5.4 Table 4: IV estimates for asset accumulation and financial market participation

Read:The IV estimates are larger than the OLS correlations. One additional year of schooling raises any investment income by 6.9 percentage points without the earned-income control and 7.5 percentage points with it. Alternative definitions of financial-market participation produce similarly large effects. The estimate for investment-income amount is about \$1,762. Schooling also raises any retirement income by 5.9 percentage points and retirement income by about \$966. In the SIPP, schooling has little effect on transaction-account ownership but raises bonds/government securities ownership by about 6.5–6.9 percentage points and stocks/mutual funds ownership by about 4 percentage points.

Economic message. The pattern is important: education does not merely make people open basic transaction accounts. It pushes households into higher-return assets and retirement/investment income. The persistence of the effect after income controls suggests that the mechanism is not simply higher wages.

5.5 Table 5: Reduced-form credit outcomes

Read:Table 5 estimates the direct effect of compulsory schooling laws on credit outcomes. Requiring schooling through grade 11 or more reduces bankruptcy by about 0.98 percentage points and foreclosure by about 1.22 percentage points relative to the eight-or-fewer-years omitted group. A one-year increase in required schooling reduces bankruptcy and foreclosure by roughly 0.2 percentage points. It also slightly increases credit scores and the fraction of balances current and reduces delinquent quarters.

Economic message. The reduced form is smaller than the structural IV effect because it averages over everyone exposed to the law, including people whose schooling was not changed by the law. Even so, stricter schooling laws are associated with better credit outcomes.

Panel A									
	(1)	(2)	(3)	(4)	(5)	(6)			
	Bankruptcy			Foreclosure					
Compulsory attendance = 9	-0.0034*			-0.0036***					
	(0.0014)			(0.0012)					
Compulsory attendance = 10	-0.013***			-0.0061***					
	(0.0027)			(0.0015)					
Compulsory attendance = 11	-0.0098***			-0.0122***					
	(0.0017)			(0.0021)					
Years of compulsory schooling		-0.0021***	-0.0019***		-0.0022***	-0.0020***			
		(0.0003)	(0.0003)		(0.0004)	(0.0003)			
No. of observations	5,750,005	5,750,005	5,750,005	5,750,005	5,750,005	5,750,005			
R ²	0.032	0.032	0.057	0.020	0.020	0.038			
p-value for F-stat of compulsory attendance	0.000			0.000					
Additional fixed effects	none	none	ZIP code	none	none	ZIP code			
Panel B									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Credit score			% Balance current			% Quarters delinquent		
Compulsory attendance = 9	-1.196**			0.0003			0.0007		
	(0.510)			(0.0004)			(0.0006)		
Compulsory attendance = 10	1.461**			0.0006			0.0002		
	(0.619)			(0.0004)			(0.0007)		
Compulsory attendance = 11	1.669**			0.0012***			-0.0021***		
	(0.680)			(0.0003)			(0.0005)		
Years of compulsory schooling		0.253**	0.056***		0.0002**	0.0001***		-0.0003**	-0.0001
		(0.106)	(0.106)		(0.0001)	(0.0001)		(0.0001)	(0.0001)
No. of observations	5,732,690	5,732,690	5,732,690	5,329,619	5,329,619	5,329,619	5,750,005	5,750,005	5,750,005
R ²	0.141	0.141	0.231	0.020	0.020	0.051	0.045	0.045	0.083
p-value for F-stat of compulsory attendance	0.000			0.003			0.000		
Additional fixed effects	none	none	ZIP code	none	none	ZIP code	none	none	ZIP code

This table reports cross-sectional regression of credit outcomes on education, measured by changes in compulsory attendance laws. The sample comprises a 9% panel of American households, restricted to borrowers who have data in every quarter of the panel from 1990 to 2011. We include 35 to 75-year-olds. Bankruptcy and Foreclosure are indicators for having undergone bankruptcy or foreclosure at least one time, respectively, between 1990 and 2011. Credit Score is averaged for each individual across all quarters of data, and is range from 280 to 850. The % of Balance Current measures the nondefault balance on credit cards divided by the total credit balance, averaged over the entire panel. The % of Quarters Delinquent is the percentage of quarters in which the borrower is delinquent on any credit card.

This table reports cross-sectional regressions of credit outcomes on education, measured by changes in compulsory attendance laws. The sample comprises a 5% panel of America borrowers, restricted to borrowers who have data in every quarter of the panel from 1999 to 2011. We include 25- to 75-year-olds. Bankruptcy and Foreclosure are indicators for having undergone bankruptcy or foreclosure at least once, respectively, between 1992 and 2011. Credit Score is averaged for each individual across all quarters of data, and it can range from 280 to 850. The % of Balance Current represents the nondelinquent balance on credit cards divided by the total credit card balance, averaged over the entire panel. The % of Quarters Delinquent

Table 6
Two-sample IV estimates of the effect of schooling on credit outcomes, FRBNY Consumer Credit Panel/Equifax

Dependent variable:	Bankruptcy	Foreclosure	Credit score	% Balance current	% Quarte delinquer
	(1)	(2)	(3)	(4)	(5)
Panel A: Entire time period					
<i>Murphy Topel standard errors</i>					
Years of schooling	-0.033*** (0.011)	-0.057*** (0.018)	7.705*** (2.781)	0.0052** (0.0023)	-0.0133* (0.0035)
No. of observations	5,198,529	5,198,529	5,182,364	4,852,175	5,198,529
<i>Bootstrap estimates</i>					
Years of schooling	-0.033 (0.023)	-0.057*** (0.010)	7.705 (4.762)	0.0052* (0.0031)	-0.0133* (0.0053)
Panel B: Pre- and postcrisis					
	1999Q2–2007Q3		2007Q3–2011Q4		
	Bankruptcy (1)	Foreclosure (2)	Bankruptcy (3)	Foreclosure (4)	
<i>Murphy Topel standard errors</i>					
Years of schooling	-0.022 (0.014)	-0.016*** (0.005)	-0.016* (0.008)	-0.045** (0.022)	
No. of observations	5,198,529	5,198,529	4,507,270	4,997,041	
<i>Bootstrap estimates</i>					
Years of schooling	-0.022 (0.024)	-0.016* (0.009)	-0.016*** (0.005)	-0.045*** (0.008)	

This table reports cross-sectional second-stage two-sample IV regressions of credit outcomes on education. sample comprises a 5% panel of American borrowers, restricted to borrowers who have data in every quarter of the panel from 1999 to 2011. We include 35- to 75-year-olds. Bankruptcy and Foreclosure are indicators having undergone bankruptcy or foreclosure at least once, respectively, between 1992 and 2011. Credit Score is averaged for each individual across all quarters of data, and it can range from 280 to 850. The % of Balance Current represents the nondelinquent balance on credit cards divided by the total credit card balance, averaged over the entire panel. The % of Quarters Delinquent represents the proportion of quarters an individual has delinquent balance on his/her credit card bills. The independent variable, years of schooling, is instrumented using compulsory schooling laws. Control variables included (coefficients not reported) in these regressions dummies for 3-year age cohorts, 10-year birth cohorts, and state-of-residence. Regressions also include state birth fixed effects interacted with a dummy variable for being born in the South and turning age 14 in 1955 later to account for the impact of *Brown v. Board of Education*. State of birth is proxied by an individual's state of residence in the first quarter of 1999. Panel A reports second-stage results for variables using all available quarters in the panel dataset. Panel B reports results for bankruptcy and foreclosure indicators separately for the precrisis period (1999Q2–2007Q3) and the postcrisis period (2007Q3–2011Q4). The top half of each p

5.6 Table 6: Two-sample IV credit outcomes

Read: Table 6 estimates the structural effect of schooling on credit outcomes using two-sample IV. In the full period, one additional year of schooling reduces bankruptcy by 3.3 percentage points and foreclosure by 5.7 percentage points, raises credit score by 7.7 points, increases the share of balances current by 0.52 percentage points, and reduces the share of delinquent quarters by 1.33 percentage points. In the period split, the foreclosure effect is about 1.6 percentage points before the crisis and about 4.5 percentage points in the crisis/postcrisis period.

Economic message. Education appears to matter most for severe financial distress. The foreclosure result is especially striking because the baseline foreclosure rate is only 5.8%. The larger crisis-period effect suggests that education helps households avoid bad outcomes when complex financial decisions become most consequential.

5.7 Mechanism evidence

Read: The paper asks whether the education effect can be explained entirely by higher labor income. The calibration assumes a 45-year-old worker earning \$20,000, saving 10% of income, earning 5% returns, and receiving a 10% wage gain from one more year of schooling. Under constant savings behavior, the added wage income generates only about \$140 of additional investment income, far below the estimated effect. To match the effect without behavioral change would require implausibly high wage returns, very high baseline savings, or very high baseline income.

Economic message. The mechanism is not only earnings. Education appears to change financial behavior: higher savings, higher-return asset participation, less negative investment income, less use of second mortgages, fewer delinquencies, fewer bankruptcies, and fewer foreclosures.

6 Short summary

- The paper studies whether education causally improves household financial outcomes.
- The main instrument is variation in state compulsory schooling laws.
- The core estimate is a LATE for people induced to obtain more schooling by these laws.
- In Census data, one more year of schooling increases any investment income by about 7–8 percentage points and raises investment-income amounts.
- In SIPP data, education raises ownership of bonds/government securities and stocks/mutual funds, but not basic transaction-account ownership.
- In FRBNY-CCP credit data, two-sample IV estimates imply large reductions in bankruptcy and foreclosure, better credit scores, and fewer delinquent credit-card outcomes.
- The foreclosure effect is especially strong during the financial crisis and its aftermath.
- The estimates remain large after controlling flexibly for earned income, suggesting that education affects financial behavior directly.
- The calibration shows that wage gains alone cannot plausibly explain the estimated increase in investment income.
- The broad conclusion is that education raises financial capability and improves household financial management, not only labor-market earnings.

7 Conclusion

Cole, Paulson, and Shastry provide causal evidence that education improves household finance. Their empirical strategy uses compulsory schooling laws to isolate variation in schooling that is not chosen by the household. The main results are large: more education increases investment and retirement-income participation, increases ownership of higher-return assets, and reduces serious credit problems such as bankruptcy and foreclosure.

The paper's most important economic message is that education affects more than wages. If additional schooling only raised labor income, the observed effect on investment income would be hard to explain. The evidence instead points to changes in savings, portfolio participation, credit management, and possibly cognitive ability or quantitative skills.

For policy, the implication is direct: education has household-finance returns that are not captured by standard wage-return calculations. More schooling can reduce costly financial mistakes and improve financial stability, especially for individuals near the dropout margin who are most affected by compulsory schooling laws.